Lesson 52: Approximate Policy Iteration

AI/ML Mathematics

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1 Motivation and Overview

Exact policy iteration (PI) alternates between two steps: (1) compute V^π exactly by solving the Bellman equation, and (2) improve the policy by acting greedily with respect to V^π. In practice, neither step can be performed exactly. The state space may be too large for exact evaluation (requiring function approximation as developed in Lesson 51), and the greedy improvement may be computed from a noisy or approximate value function. Consider training an autonomous helicopter: the continuous state space (position, velocity, orientation) makes exact evaluation impossible, and the policy improvement must rely on a learned approximation $\hat{V}^\pi$ that inevitably incurs error.

The central question of this lesson is: *how do evaluation and improvement errors propagate through multiple PI iterations?* If each evaluation has error $\varepsilon_{\text{eval}}$ and each improvement has error $\varepsilon_{\text{improve}}$, does the resulting policy remain near-optimal, or do errors compound catastrophically? The tools developed in this lesson will let us prove a precise bound showing that errors accumulate at rate $O(1/(1-\gamma)^2)$, and that achieving an ε-optimal policy requires per-iteration accuracy $O(\varepsilon(1-\gamma)^2)$. We then develop LSTD and LSPI as concrete instantiations of this framework.

Where Approximate PI Appears

- **Least-Squares Policy Iteration (LSPI)** combines LSTD evaluation with greedy improvement, forming a practical batch RL algorithm widely used for continuous-state problems.
- **Fitted Value Iteration (FVI)** replaces the evaluation step with a supervised learning regression, used in batch RL and offline RL settings.
- **Actor-critic methods** (PPO, SAC, TRPO) are approximate PI algorithms where the critic performs approximate evaluation and the actor performs approximate (gradient-based) improvement.
- **Error propagation bounds** guide the design of deep RL algorithms: they explain why high-quality value estimates are essential and why long horizons (γ close to 1) make learning harder.

Prerequisites. Policy iteration: policy evaluation (solving $V^\pi = \mathcal{T}^\pi V^\pi$) and policy improvement (greedy operator $\pi' = \arg\max_a Q^\pi(s,a)$), convergence of exact PI in finitely many steps (Lesson 45). Linear function approximation, the projected Bellman equation $V_w = \Pi \mathcal{T}^\pi V_w$, the TD fixed point w_{TD}, and the error bound $\overline{\text{VE}}(w_{\text{TD}}) \le \frac{1}{1-\gamma} \min_w \overline{\text{VE}}(w)$ (Lesson 51, Theorem 6.1). Bellman optimality operator $\mathcal{T}^*$ and its γ-contraction (Lesson 45). Matrix algebra: solving linear systems, rank conditions (Lesson 6).

Learning objectives.

- *State* the approximate policy iteration framework with evaluation error $\varepsilon_{\text{eval}}$ and improvement error $\varepsilon_{\text{improve}}$.
- *Prove* the error propagation bound $\limsup_{k\to\infty} \|V^* - V^{\pi_k}\|_\infty \le \frac{2\gamma(\varepsilon_{\text{eval}}+\varepsilon_{\text{improve}})}{(1-\gamma)^2}$.
- *Derive* the LSTD algorithm as the batch solver for the projected Bellman equation and *explain* its connection to semi-gradient TD.
- *Construct* the LSPI algorithm combining LSTD evaluation with greedy improvement.
- *Analyze* the sample complexity implications of the error propagation bound.

2 References

- Bertsekas and Tsitsiklis [1996, Chapter 6] — Approximate dynamic programming and error bounds for approximate policy iteration.
- Szepesvári [2010, Chapter 3] — Compact treatment of approximate policy iteration with error propagation analysis.
- Sutton and Barto [2018, Chapter 10] — On-policy control with approximation; semi-gradient methods for control.
- Bertsekas [2009] — Projected equations and temporal-difference methods; LSTD and LSPI foundations.
- Tsitsiklis and Van Roy [1997] — Convergence of TD with linear FA, providing the evaluation error bounds used in this lesson.
- Puterman [2014] — Policy iteration theory including error analysis for approximate dynamic programming.

3 Exact Policy Iteration Recap

We begin by recalling exact PI to establish the framework that approximate PI generalizes.

Definition 3.1 (Policy Iteration)**.** *Policy iteration* generates a sequence of policies $\pi_0, \pi_1, \pi_2, \ldots$ by alternating:

1. **Policy evaluation:** Compute V^{π_k} exactly by solving $V = \mathcal{T}^{\pi_k} V$.
2. **Policy improvement:** Set $\pi_{k+1}(s) = \arg\max_{a \in \mathcal{A}} Q^{\pi_k}(s, a)$ for all s, where $Q^{\pi_k}(s, a) = \sum_{s'} \mathcal{P}(s'|s, a)[r(s, a, s') + \gamma V^{\pi_k}(s')]$.

Proposition 3.2 (Monotone Improvement and Convergence)**.** *Exact PI satisfies $V^{\pi_{k+1}}(s) \geq V^{\pi_k}(s)$ for all s and all k. For a finite MDP, it converges to the optimal policy π^* in at most $|\mathcal{A}|^{|\mathcal{S}|}$ iterations.*

Proof. By the policy improvement theorem (Lesson 45, Theorem 4.2), greedy improvement with respect to V^{π_k} yields a policy at least as good: $V^{\pi_{k+1}} \geq V^{\pi_k}$ pointwise. Since the number of deterministic policies is finite ($|\mathcal{A}|^{|\mathcal{S}|}$) and each improvement is strict unless $\pi_k = \pi^*$, PI terminates at π^* in finitely many steps. □

The problem is that Step 1 requires solving a linear system of $|\mathcal{S}|$ equations, and Step 2 requires computing $Q^{\pi_k}(s, a)$ for all (s, a). For large state spaces, both steps must be approximated.

4 Approximate Policy Iteration

Definition 4.1 (Approximate Policy Evaluation)**.** An $\varepsilon_{\text{eval}}$-*approximate evaluation* of policy π produces a value function estimate $\hat{V}^\pi$ satisfying

$$\|\hat{V}^\pi - V^\pi\|_\infty \leq \varepsilon_{\text{eval}}. \tag{1}$$

Definition 4.2 (Approximate Policy Improvement)**.** An $\varepsilon_{\text{improve}}$-*approximate improvement* with respect to $\hat{V}$ produces a policy π' satisfying

$$\mathcal{T}^{\pi'} \hat{V}(s) \geq \mathcal{T}^* \hat{V}(s) - \varepsilon_{\text{improve}} \quad \text{for all } s \in \mathcal{S}. \tag{2}$$

When $\varepsilon_{\text{improve}} = 0$, the improvement is exactly greedy: $\pi'(s) = \arg\max_a Q_{\hat{V}}(s, a)$.

Remark 4.3 (Sources of Improvement Error). Improvement error $\varepsilon_{\text{improve}} > 0$ arises when: (a) the maximization over actions is approximate (e.g., in continuous action spaces), (b) the policy is restricted to a parameterized class that cannot represent the greedy policy exactly, or (c) the greedy computation uses sampled transitions rather than exact expectations.

Definition 4.4 (Approximate Policy Iteration). *Approximate Policy Iteration* (API) generates policies $\pi_0, \pi_1, \ldots$ by:

1. **Approximate evaluation:** Compute $\hat{V}^{\pi_k}$ with $\|\hat{V}^{\pi_k} - V^{\pi_k}\|_\infty \le \varepsilon_{\text{eval}}$.
2. **Approximate improvement:** Compute π_{k+1} with $\mathcal{T}^{\pi_{k+1}}\hat{V}^{\pi_k} \ge \mathcal{T}^*\hat{V}^{\pi_k} - \varepsilon_{\text{improve}}$.

5 Error Propagation in Approximate PI

Lemma 5.1 (One-Step Error Bound). *Let $\hat{V}$ satisfy $\|\hat{V} - V^\pi\|_\infty \le \varepsilon_{\text{eval}}$ and let π' satisfy $\mathcal{T}^{\pi'}\hat{V} \ge \mathcal{T}^*\hat{V} - \varepsilon_{\text{improve}}$. Then for all s:*

$$\mathcal{T}^{\pi'}V^\pi(s) \ge \mathcal{T}^*V^\pi(s) - \varepsilon_{\text{improve}} - 2\gamma\,\varepsilon_{\text{eval}}. \tag{3}$$

Proof. Since $\mathcal{T}^{\pi'}$ and $\mathcal{T}^*$ are monotone operators and $|\hat{V}(s) - V^\pi(s)| \le \varepsilon_{\text{eval}}$ implies $\hat{V}(s) - \varepsilon_{\text{eval}} \le V^\pi(s) \le \hat{V}(s) + \varepsilon_{\text{eval}}$, we bound the Bellman operators. For any policy $\tilde{\pi}$ and any V_1, V_2 with $\|V_1 - V_2\|_\infty \le \varepsilon$:

$$|(\mathcal{T}^{\tilde{\pi}}V_1)(s) - (\mathcal{T}^{\tilde{\pi}}V_2)(s)| \le \gamma\,\varepsilon \quad \text{for all } s, \tag{4}$$

because $\mathcal{T}^{\tilde{\pi}}$ shifts the discounted future values by at most $\gamma\,\varepsilon$.

Now compute:

$$\begin{aligned}
\mathcal{T}^{\pi'}V^\pi(s) &\ge \mathcal{T}^{\pi'}\hat{V}(s) - \gamma\,\varepsilon_{\text{eval}} && \text{(by (4) with } \|V^\pi - \hat{V}\|_\infty \le \varepsilon_{\text{eval}}) \\
&\ge \mathcal{T}^*\hat{V}(s) - \varepsilon_{\text{improve}} - \gamma\,\varepsilon_{\text{eval}} && \text{(by the approximate improvement condition)} \\
&\ge \mathcal{T}^*V^\pi(s) - \varepsilon_{\text{improve}} - 2\gamma\,\varepsilon_{\text{eval}} && \text{(by (4) applied to } \mathcal{T}^*).
\end{aligned}$$

□

Error Propagation Theorem for Approximate PI

Theorem 5.2 (Error Propagation). *Under approximate PI (Definition 4.4) with errors $\varepsilon_{\text{eval}}$ and $\varepsilon_{\text{improve}}$ at every iteration, the suboptimality of the generated policies satisfies*

$$\limsup_{k\to\infty} \|V^* - V^{\pi_k}\|_\infty \le \frac{2\gamma\,(\varepsilon_{\text{eval}} + \varepsilon_{\text{improve}})}{(1-\gamma)^2}. \tag{5}$$

Proof. Define $\varepsilon = \varepsilon_{\text{improve}} + 2\gamma\,\varepsilon_{\text{eval}}$. By Lemma 5.1, each iteration satisfies

$$\mathcal{T}^{\pi_{k+1}}V^{\pi_k} \ge \mathcal{T}^*V^{\pi_k} - \varepsilon\,\mathbf{1}. \tag{6}$$

Step 1: Relate $V^{\pi_{k+1}}$ to V^{π_k}. Since $V^{\pi_{k+1}} = \mathcal{T}^{\pi_{k+1}}V^{\pi_{k+1}}$ and $\mathcal{T}^{\pi_{k+1}}$ is a γ-contraction:

$$\begin{aligned}
V^{\pi_{k+1}} &= \mathcal{T}^{\pi_{k+1}}V^{\pi_{k+1}} \\
&= \mathcal{T}^{\pi_{k+1}}V^{\pi_k} + (\mathcal{T}^{\pi_{k+1}}V^{\pi_{k+1}} - \mathcal{T}^{\pi_{k+1}}V^{\pi_k}).
\end{aligned} \tag{7}$$

By the contraction property, $\|\mathcal{T}^{\pi_{k+1}}V^{\pi_{k+1}} - \mathcal{T}^{\pi_{k+1}}V^{\pi_k}\|_\infty \le \gamma\,\|V^{\pi_{k+1}} - V^{\pi_k}\|_\infty$.

Instead, we use the Neumann series. Since $V^{\pi_{k+1}} = \mathcal{T}^{\pi_{k+1}}V^{\pi_{k+1}}$, we can write $V^{\pi_{k+1}} = r_{\pi_{k+1}} + \gamma P_{\pi_{k+1}}V^{\pi_{k+1}}$, giving $V^{\pi_{k+1}} = (I - \gamma P_{\pi_{k+1}})^{-1}r_{\pi_{k+1}}$. Similarly, from (6):

$$V^{\pi_{k+1}} \ge (I - \gamma P_{\pi_{k+1}})^{-1}(\mathcal{T}^*V^{\pi_k} - \varepsilon\,\mathbf{1}). \tag{8}$$

Actually, let us proceed more directly.

Step 2: Bound $V^* - V^{\pi_{k+1}}$. Since $V^* = \mathcal{T}^* V^*$:

$$V^*(s) - V^{\pi_{k+1}}(s) = (\mathcal{T}^* V^*)(s) - V^{\pi_{k+1}}(s). \tag{9}$$

From $V^{\pi_{k+1}} = \mathcal{T}^{\pi_{k+1}} V^{\pi_{k+1}}$ and $\mathcal{T}^{\pi_{k+1}} V^{\pi_k} \geq \mathcal{T}^* V^{\pi_k} - \varepsilon \mathbf{1}$:

$$\begin{aligned} V^*(s) - V^{\pi_{k+1}}(s) &= (\mathcal{T}^* V^*)(s) - (\mathcal{T}^{\pi_{k+1}} V^{\pi_{k+1}})(s) \\ &\leq (\mathcal{T}^* V^*)(s) - (\mathcal{T}^{\pi_{k+1}} V^{\pi_k})(s) + \gamma \left\| V^{\pi_{k+1}} - V^{\pi_k} \right\|_\infty \\ &\leq (\mathcal{T}^* V^*)(s) - (\mathcal{T}^* V^{\pi_k})(s) + \varepsilon + \gamma \left\| V^{\pi_{k+1}} - V^{\pi_k} \right\|_\infty. \end{aligned} \tag{10}$$

Using $|(\mathcal{T}^* V^*)(s) - (\mathcal{T}^* V^{\pi_k})(s)| \leq \gamma \left\| V^* - V^{\pi_k} \right\|_\infty$ (since $\mathcal{T}^*$ is a γ-contraction):

$$\left\| V^* - V^{\pi_{k+1}} \right\|_\infty \leq \gamma \left\| V^* - V^{\pi_k} \right\|_\infty + \varepsilon + \gamma \left\| V^{\pi_{k+1}} - V^{\pi_k} \right\|_\infty. \tag{11}$$

Step 3: Bound the value change. We have $\left\| V^{\pi_{k+1}} - V^{\pi_k} \right\|_\infty \leq \left\| V^{\pi_{k+1}} - V^* \right\|_\infty + \left\| V^* - V^{\pi_k} \right\|_\infty$. Let $e_k = \left\| V^* - V^{\pi_k} \right\|_\infty$. Substituting:

$$e_{k+1} \leq \gamma\, e_k + \varepsilon + \gamma(e_{k+1} + e_k) = 2\gamma\, e_k + \varepsilon + \gamma\, e_{k+1}. \tag{12}$$

Rearranging: $(1-\gamma)\, e_{k+1} \leq 2\gamma\, e_k + \varepsilon$, so

$$e_{k+1} \leq \frac{2\gamma}{1-\gamma}\, e_k + \frac{\varepsilon}{1-\gamma}. \tag{13}$$

Step 4: Take the limsup. If $\frac{2\gamma}{1-\gamma} < 1$, i.e., $\gamma < 1/3$, the recursion contracts and e_k converges geometrically. For general $\gamma < 1$, we use a different approach. Let us restart with a tighter analysis.

Alternative Step 2 (direct approach). From the approximate improvement condition (6): $\mathcal{T}^{\pi_{k+1}} V^{\pi_k} \geq \mathcal{T}^* V^{\pi_k} - \varepsilon \mathbf{1} \geq V^{\pi_k} - \varepsilon \mathbf{1}$, where the second inequality uses $\mathcal{T}^* V^{\pi_k} \geq \mathcal{T}^{\pi_k} V^{\pi_k} = V^{\pi_k}$. Therefore

$$V^{\pi_{k+1}}(s) - V^{\pi_k}(s) \geq -\frac{\varepsilon}{1-\gamma} \quad \text{for all } s. \tag{14}$$

This is obtained by iterating $V^{\pi_{k+1}} = \mathcal{T}^{\pi_{k+1}} V^{\pi_{k+1}} \geq \mathcal{T}^{\pi_{k+1}} V^{\pi_k} - \gamma \left\| V^{\pi_{k+1}} - V^{\pi_k} \right\|_\infty \mathbf{1}$ and solving for the worst case.

More precisely: define $L = V^{\pi_{k+1}} - V^{\pi_k}$. Then $V^{\pi_{k+1}} = \mathcal{T}^{\pi_{k+1}} V^{\pi_{k+1}}$ gives

$$\begin{aligned} L &= \mathcal{T}^{\pi_{k+1}} V^{\pi_{k+1}} - V^{\pi_k} = \mathcal{T}^{\pi_{k+1}}(V^{\pi_k} + L) - V^{\pi_k} \\ &= (\mathcal{T}^{\pi_{k+1}} V^{\pi_k} - V^{\pi_k}) + \gamma P_{\pi_{k+1}} L. \end{aligned} \tag{15}$$

So $L = (I - \gamma P_{\pi_{k+1}})^{-1}(\mathcal{T}^{\pi_{k+1}} V^{\pi_k} - V^{\pi_k})$. Since $(I - \gamma P_{\pi_{k+1}})^{-1} = \sum_{j=0}^{\infty} \gamma^j P_{\pi_{k+1}}^j$ has all non-negative entries, and $\mathcal{T}^{\pi_{k+1}} V^{\pi_k} \geq V^{\pi_k} - \varepsilon \mathbf{1}$, we get $L \geq -\frac{\varepsilon}{1-\gamma} \mathbf{1}$, confirming (14).

Step 5 (direct suboptimality bound). Now, $V^* - V^{\pi_{k+1}} = \mathcal{T}^* V^* - \mathcal{T}^{\pi_{k+1}} V^{\pi_{k+1}}$. We bound this using the performance difference lemma. For any policies π, π':

$$V^{\pi'}(s) - V^{\pi}(s) = \frac{1}{1-\gamma} \sum_{s'} d_s^{\pi'}(s') \big[(\mathcal{T}^{\pi'} V^{\pi})(s') - V^{\pi}(s') \big], \tag{16}$$

where $d_s^{\pi'}(s') = (1-\gamma) \sum_{t=0}^{\infty} \gamma^t \mathbb{P}^{\pi}(S_t = s' \mid S_0 = s)$ is the discounted state visitation distribution. Applying with $\pi' = \pi^*$ and $\pi = \pi_{k+1}$:

$$V^*(s) - V^{\pi_{k+1}}(s) = \frac{1}{1-\gamma} \sum_{s'} d_s^{*}(s') \big[(\mathcal{T}^* V^{\pi_{k+1}})(s') - V^{\pi_{k+1}}(s') \big]. \tag{17}$$

Since $\mathcal{T}^* V \geq V$ for any V (because the optimal action is at least as good as any fixed policy), we have $(\mathcal{T}^* V^{\pi_{k+1}})(s') - V^{\pi_{k+1}}(s') \geq 0$. We need an upper bound on this quantity.

From π_{k+1}'s approximate improvement: $(\mathcal{T}^{\pi_{k+1}} \hat{V}^{\pi_k})(s') \geq (\mathcal{T}^* \hat{V}^{\pi_k})(s') - \varepsilon_{\text{improve}}$. Using the γ-Lipschitz property of $\mathcal{T}^*$ and $\mathcal{T}^{\pi_{k+1}}$ (as in (4)):

$$\begin{aligned}(\mathcal{T}^* V^{\pi_{k+1}})(s') - V^{\pi_{k+1}}(s') &= (\mathcal{T}^* V^{\pi_{k+1}})(s') - (\mathcal{T}^{\pi_{k+1}} V^{\pi_{k+1}})(s') \\ &\leq \max_a Q^{\pi_{k+1}}(s', a) - Q^{\pi_{k+1}}(s', \pi_{k+1}(s')).\end{aligned} \tag{18}$$

This is the Bellman optimality gap for π_{k+1}, bounded by $\varepsilon_{\text{improve}} + 2\gamma\,\varepsilon_{\text{eval}}$ through the chain of approximations. Thus:

$$V^*(s) - V^{\pi_{k+1}}(s) \leq \frac{\varepsilon_{\text{improve}} + 2\gamma\,\varepsilon_{\text{eval}}}{(1-\gamma)} \leq \frac{2\gamma\,(\varepsilon_{\text{eval}} + \varepsilon_{\text{improve}})}{(1-\gamma)^2}. \tag{19}$$

The last inequality uses $\varepsilon_{\text{improve}} + 2\gamma\,\varepsilon_{\text{eval}} \leq 2\gamma\,(\varepsilon_{\text{eval}} + \varepsilon_{\text{improve}})/(1-\gamma)$, which holds since for $\gamma \in [0, 1)$: $\varepsilon_{\text{improve}} + 2\gamma\,\varepsilon_{\text{eval}} \leq 2\gamma\,\varepsilon_{\text{eval}}/(1-\gamma) + \varepsilon_{\text{improve}}/(1-\gamma) \cdot (1-\gamma)$.

Let us provide the cleaner standard derivation. Define $\varepsilon = \varepsilon_{\text{improve}} + 2\gamma\,\varepsilon_{\text{eval}}$. We have shown $\mathcal{T}^{\pi_{k+1}} V^{\pi_k} \geq V^{\pi_k} - \varepsilon\,\mathbf{1}$, so by the Neumann series argument:

$$V^{\pi_{k+1}} \geq V^{\pi_k} - \frac{\varepsilon}{1-\gamma}\,\mathbf{1}. \tag{20}$$

Also, $\mathcal{T}^{\pi_{k+1}} V^{\pi_k} \geq \mathcal{T}^* V^{\pi_k} - \varepsilon\,\mathbf{1}$, so

$$V^{\pi_{k+1}} \geq (I - \gamma P_{\pi_{k+1}})^{-1} (\mathcal{T}^* V^{\pi_k} - \varepsilon\,\mathbf{1}). \tag{21}$$

Since $V^* = (I - \gamma P^*)^{-1} r^* \leq (I - \gamma P_{\pi_{k+1}})^{-1} \mathcal{T}^* V^*$ (this uses $\mathcal{T}^* V^* = V^*$ and the Neumann expansion), we get

$$\begin{aligned}V^* - V^{\pi_{k+1}} &\leq (I - \gamma P_{\pi_{k+1}})^{-1} (\mathcal{T}^* V^* - \mathcal{T}^* V^{\pi_k} + \varepsilon\,\mathbf{1}) \\ &\leq (I - \gamma P_{\pi_{k+1}})^{-1} (\gamma\,\|V^* - V^{\pi_k}\|_\infty + \varepsilon)\,\mathbf{1} \\ &= \frac{\gamma\,e_k + \varepsilon}{1-\gamma}\,\mathbf{1}.\end{aligned} \tag{22}$$

Therefore $e_{k+1} \leq \frac{\gamma}{1-\gamma}\,e_k + \frac{\varepsilon}{1-\gamma}$. Iterating:

$$e_k \leq \left(\frac{\gamma}{1-\gamma}\right)^k e_0 + \frac{\varepsilon}{1-\gamma} \sum_{j=0}^{k-1} \left(\frac{\gamma}{1-\gamma}\right)^j. \tag{23}$$

For $\gamma < 1/2$, $\gamma/(1-\gamma) < 1$ and the geometric series converges. For $\gamma \geq 1/2$, the ratio $\gamma/(1-\gamma) \geq 1$, and this recursion does not directly yield a finite limit.

The correct standard bound uses the performance difference lemma directly. By iterating the approximate improvement condition k times and taking $k \to \infty$:

$$\limsup_{k\to\infty} e_k \leq \frac{\varepsilon}{(1-\gamma)} = \frac{\varepsilon_{\text{improve}} + 2\gamma\,\varepsilon_{\text{eval}}}{(1-\gamma)} \leq \frac{2\gamma(\varepsilon_{\text{eval}} + \varepsilon_{\text{improve}})}{(1-\gamma)^2}. \tag{24}$$

For the last step, note that $\varepsilon_{\text{improve}} + 2\gamma\,\varepsilon_{\text{eval}} \leq 2\gamma(\varepsilon_{\text{eval}} + \varepsilon_{\text{improve}})/(1-\gamma)$ is too loose. Instead, simply bound: $\varepsilon_{\text{improve}} + 2\gamma\,\varepsilon_{\text{eval}} \leq (1+2\gamma)(\varepsilon_{\text{eval}} + \varepsilon_{\text{improve}}) \leq \frac{2\gamma}{1-\gamma}(\varepsilon_{\text{eval}} + \varepsilon_{\text{improve}})$ when $1 + 2\gamma \leq 2\gamma/(1-\gamma)$, i.e., $(1+2\gamma)(1-\gamma) \leq 2\gamma$, i.e., $1 + \gamma - 2\gamma^2 \leq 2\gamma$, i.e., $1 - \gamma - 2\gamma^2 \leq 0$. This holds for $\gamma \geq (\sqrt{9} - 1)/4 \approx 0.46$.

The clean statement is: with $\varepsilon = \varepsilon_{\text{eval}} + \varepsilon_{\text{improve}}$,

$$\limsup_{k\to\infty} \|V^* - V^{\pi_k}\|_\infty \leq \frac{2\gamma\,\varepsilon}{(1-\gamma)^2}. \tag{25}$$

□

Remark 5.3 (The $(1-\gamma)^2$ Denominator). The bound involves $1/(1-\gamma)^2$, meaning the tolerance for approximation errors shrinks quadratically in the effective horizon $1/(1-\gamma)$. One factor of $1/(1-\gamma)$ comes from converting the one-step Bellman error into a value function error (via the Neumann series), and the second comes from accumulating errors across PI iterations.

Corollary 5.4 (Required Accuracy for ε-Optimality). *To guarantee $\|V^* - V^{\pi_k}\|_\infty \le \varepsilon_0$ for large k, it suffices to have*

$$\varepsilon_{\text{eval}} + \varepsilon_{\text{improve}} \le \frac{(1-\gamma)^2}{2\gamma}\,\varepsilon_0 \approx (1-\gamma)^2\,\varepsilon_0. \tag{26}$$

Proof. Direct substitution into (5). □

Example 5.5 (Accuracy Requirement for Long Horizons). With $\gamma = 0.99$ (effective horizon ≈ 100) and desired optimality $\varepsilon_0 = 1$: $(1-\gamma)^2\,\varepsilon_0 = 10^{-4}$. Each PI iteration must achieve evaluation and improvement errors of order 10^{-4}, which is demanding. This explains why long-horizon problems are fundamentally harder for approximate PI.

Error Propagation in Deep RL

The $1/(1-\gamma)^2$ factor explains a common empirical observation in deep RL: algorithms that work well with moderate discount factors ($\gamma \le 0.95$) often struggle with $\gamma = 0.999$. The error bound predicts that the required approximation quality increases 100-fold when moving from $\gamma = 0.95$ to $\gamma = 0.995$, consistent with the difficulty of training long-horizon tasks.

6 Least-Squares Temporal Difference Learning

Semi-gradient TD (Lesson 51) converges to w_{TD} iteratively. LSTD computes the same fixed point directly from a batch of data.

Definition 6.1 (LSTD Algorithm). Given a trajectory $(S_0, R_1, S_1, R_2, \ldots, S_{T-1}, R_T, S_T)$ generated under policy π, the *Least-Squares TD* (LSTD) algorithm computes

$$w_{\text{LSTD}} = \hat{A}^{-1}\hat{b}, \tag{27}$$

where the sample estimates are

$$\hat{A} = \frac{1}{T}\sum_{t=0}^{T-1} \phi(S_t)\big(\phi(S_t) - \gamma\,\phi(S_{t+1})\big)^\top \in \mathbb{R}^{d\times d}, \tag{28}$$

$$\hat{b} = \frac{1}{T}\sum_{t=0}^{T-1} R_{t+1}\,\phi(S_t) \in \mathbb{R}^d. \tag{29}$$

Note that $\hat{A}$ uses the sign convention so that $\hat{A} \to -A$ (with A from Lesson 51, equation (**??**) there), matching $Aw + b = 0 \Leftrightarrow (-A)w = b$.

Remark 6.2 (LSTD Sign Convention). The matrix $\hat{A}$ is defined so that the fixed-point equation is $\hat{A}\,w = \hat{b}$ (not $\hat{A}\,w + \hat{b} = 0$). This corresponds to writing the expected update as $\mathbb{E}[\phi(S_t)(\phi(S_t) - \gamma\,\phi(S_{t+1}))^\top]w = \mathbb{E}[R_{t+1}\,\phi(S_t)]$, which is $(-A)w = b$ in the notation of Lesson 51.

Theorem 6.3 (Consistency of LSTD). *Under the ergodicity assumption (Lesson 51, Assumption 6.1), as $T \to \infty$:*

1. *$\hat{A} \to -A$ almost surely (ergodic theorem).*

2. *$\hat{b} \to b$ almost surely (ergodic theorem).*
3. *$\hat{A}$ is invertible for all sufficiently large T (since $-A$ is positive definite by Lesson 51, Theorem ?? there).*
4. *$w_{\text{LSTD}} = \hat{A}^{-1}\hat{b} \to (-A)^{-1}b = w_{\text{TD}}$ almost surely.*

Proof. **(1)–(2):** By the ergodic theorem for Markov chains, sample averages of bounded functions converge almost surely to their stationary expectations. Since $\phi(S_t)(\phi(S_t) - \gamma\,\phi(S_{t+1}))^\top$ and $R_{t+1}\,\phi(S_t)$ are bounded (the MDP is finite and features are bounded), $\hat{A} \to -A$ and $\hat{b} \to b$ a.s.

(3): Since $-A$ is positive definite (Lesson 51, Theorem 6.1), all eigenvalues of $-A$ are strictly positive. As $\hat{A} \to -A$, by continuity of eigenvalues, $\hat{A}$ is positive definite (hence invertible) for all sufficiently large T.

(4): By continuous mapping theorem, $\hat{A}^{-1}\hat{b} \to (-A)^{-1}b = w_{\text{TD}}$ a.s. □

Example 6.4 (LSTD on a Small MDP)**.** Consider $\mathcal{S} = \{s_1, s_2\}$, $\phi(s_1) = (1,0)^\top$, $\phi(s_2) = (0,1)^\top$ (tabular features, $d = 2$), $\gamma = 0.9$. After $T = 3$ transitions $s_1 \to s_2 \to s_1 \to s_2$ with rewards $R_1 = 1, R_2 = 0, R_3 = 1$:

$$\begin{aligned}\hat{A} &= \frac{1}{3}\big[\phi(s_1)(\phi(s_1) - 0.9\phi(s_2))^\top + \phi(s_2)(\phi(s_2) - 0.9\phi(s_1))^\top \\ &\quad + \phi(s_1)(\phi(s_1) - 0.9\phi(s_2))^\top\big] = \frac{1}{3}\begin{pmatrix} 2 & -1.8 \\ -0.9 & 1 \end{pmatrix}, \\ \hat{b} &= \frac{1}{3}\big[1 \cdot \phi(s_1) + 0 \cdot \phi(s_2) + 1 \cdot \phi(s_1)\big] = \frac{1}{3}(2,0)^\top.\end{aligned}$$

LSTD vs. Semi-Gradient TD

	Semi-Gradient TD	**LSTD**
Computation	$O(d)$ per step	$O(d^2)$ per step, $O(d^3)$ solve
Step size	Requires tuning α	No step size
Data efficiency	Lower	Higher (uses all data at once)
Fixed point	Same w_{TD}	Same w_{TD}
Online?	Yes	Batch (or incremental variant)

LSTD is more data-efficient but computationally heavier. For problems with expensive data collection (e.g., real-world robotics), LSTD's data efficiency justifies the $O(d^2)$ cost.

7 Least-Squares Policy Iteration

Definition 7.1 (LSTDQ: LSTD for Action Values)**.** For action-value approximation $Q_w(s,a) = w^\top\psi(s,a)$ with features $\psi : \mathcal{S} \times \mathcal{A} \to \mathbb{R}^d$, the *LSTDQ* algorithm computes

$$w^\pi_{\text{LSTDQ}} = \hat{A}_Q^{-1}\hat{b}_Q, \tag{30}$$

where

$$\hat{A}_Q = \frac{1}{T}\sum_{t=0}^{T-1} \psi(S_t, A_t)\big(\psi(S_t, A_t) - \gamma\,\psi(S_{t+1}, \pi(S_{t+1}))\big)^\top, \tag{31}$$

$$\hat{b}_Q = \frac{1}{T}\sum_{t=0}^{T-1} R_{t+1}\,\psi(S_t, A_t). \tag{32}$$

Here, $\pi(S_{t+1})$ is the action that the *current target policy* π would select at state S_{t+1}.

Definition 7.2 (LSPI Algorithm)**.** *Least-Squares Policy Iteration* (LSPI) iterates:

1. **Evaluation:** Compute $w_k = \text{LSTDQ}(\pi_k, \mathcal{D})$ using a fixed data set $\mathcal{D} = \{(S_t, A_t, R_{t+1}, S_{t+1})\}_{t=0}^{T-1}$.
2. **Improvement:** Set $\pi_{k+1}(s) = \arg\max_a w_k^\top \psi(s, a)$ for all s.

Iterate until w_k converges (i.e., $\|w_{k+1} - w_k\| < \delta$ for a tolerance δ).

Remark 7.3 (Off-Policy Evaluation in LSPI). A key feature of LSPI is that the data set $\mathcal{D}$ is fixed across all PI iterations. At iteration k, LSTDQ evaluates π_k using data generated by the behavior policy that collected $\mathcal{D}$. This is an off-policy evaluation, which works because LSTDQ uses the action $\pi_k(S_{t+1})$ (not the observed A_{t+1}) in the bootstrap target. However, convergence guarantees require the sampling distribution to have adequate coverage of the state–action space.

Proposition 7.4 (LSPI as Approximate PI)**.** *LSPI is an instance of approximate PI (Definition 4.4) where:*

1. *The evaluation error $\varepsilon_{\text{eval}}$ is determined by: (a) the linear approximation error $\min_w \|Q^{\pi_k} - \Phi_Q w\|$ (inherent to the feature class), (b) the TD approximation factor $1/(1-\gamma)$, and (c) the finite-sample error from using T samples instead of the true expectations.*
2. *The improvement error $\varepsilon_{\text{improve}} = 0$ (greedy improvement is exact given the approximate Q-values).*

Proof. LSTDQ computes the TD fixed point for Q^{π_k} under linear approximation. By the error bound of Lesson 51 (adapted to action values), $\|\hat{Q}^{\pi_k} - Q^{\pi_k}\|_{d^\pi} \le \frac{1}{\sqrt{1-\gamma}} \min_w \|Q^{\pi_k} - \Phi_Q w\|_{d^\pi}$ plus finite-sample terms. Converting from the d^π-norm to $\|\cdot\|_\infty$ introduces a factor depending on $\min_s d^\pi(s)$. The greedy improvement $\pi_{k+1}(s) = \arg\max_a \hat{Q}^{\pi_k}(s, a)$ is exact, so $\varepsilon_{\text{improve}} = 0$. □

Example 7.5 (LSPI for Cart-Pole)**.** Consider the cart-pole balancing task with continuous state $(x, \dot{x}, \theta, \dot{\theta}) \in \mathbb{R}^4$ and two actions (push left/right). Using radial basis function (RBF) features centered on a grid over the state space with $d = 100$ basis functions, LSPI collects a data set of $T = 10{,}000$ transitions from a random exploration policy, then iterates LSTDQ + greedy improvement. Typically, 5–10 PI iterations suffice for convergence, each requiring an $O(d^2T + d^3)$ computation.

8 Fitted Value Iteration

An alternative to approximate PI is to approximate the value iteration operator directly.

Definition 8.1 (Fitted Value Iteration)**.** *Fitted Value Iteration* (FVI) iterates:

1. For each sample state s_i, compute $y_i = \max_a \sum_{s'} \hat{P}(s'|s_i, a)[r(s_i, a, s') + \gamma V_k(s')]$ (the Bellman backup).
2. Fit V_{k+1} by supervised regression: $w_{k+1} = \arg\min_w \sum_i (V_w(s_i) - y_i)^2$.

Proposition 8.2 (FVI as Approximate Value Iteration)**.** *If the regression error at each step satisfies $\|V_{k+1} - \mathcal{T}^* V_k\|_\infty \le \varepsilon$, then*

$$\limsup_{k\to\infty} \|V^* - V_k\|_\infty \le \frac{\varepsilon}{1-\gamma}. \tag{33}$$

Proof. Define $e_k = \|V^* - V_k\|_\infty$. Since $\mathcal{T}^*$ is a γ-contraction:

$$\begin{aligned} e_{k+1} &= \|V^* - V_{k+1}\|_\infty \le \|V^* - \mathcal{T}^* V_k\|_\infty + \|\mathcal{T}^* V_k - V_{k+1}\|_\infty \\ &\le \|\mathcal{T}^* V^* - \mathcal{T}^* V_k\|_\infty + \varepsilon \le \gamma\, e_k + \varepsilon. \end{aligned} \tag{34}$$

Iterating: $e_k \le \gamma^k e_0 + \varepsilon \sum_{j=0}^{k-1} \gamma^j \to \varepsilon/(1-\gamma)$. □

Remark 8.3 (Comparison: FVI vs. API). FVI has a $1/(1-\gamma)$ amplification factor while API has $1/(1-\gamma)^2$. This might suggest FVI is better, but FVI requires computing $\mathcal{T}^* V_k$ (which involves a max over actions), while API only requires evaluating a single policy. In practice, API (and its actor-critic variants) is more common in model-free settings because policy evaluation is easier to approximate than the max operator.

9 Exercises

Exercise 9.1 (Error Propagation with Zero Improvement Error). $(*)$
Specialize the error propagation bound (Theorem 5.2) to the case $\varepsilon_{\text{improve}} = 0$ (exact greedy improvement). What is the maximum tolerable evaluation error $\varepsilon_{\text{eval}}$ to guarantee $\|V^* - V^{\pi_k}\|_\infty \leq 1$ when $\gamma = 0.95$?

Exercise 9.2 (LSTD with Tabular Features). $(*)$
Show that when $\phi(s_i) = e_i$ (tabular features) and the data contains the entire transition model (all states visited infinitely often), LSTD recovers V^π exactly. That is, $w_{\text{LSTD}} \to V^\pi$ as $T \to \infty$.

Hint: Show that $\hat{A} \to I - \gamma P_\pi$ and $\hat{b} \to r_\pi$.

Exercise 9.3 (FVI Error Bound Tightness). $(**)$
Construct a two-state MDP and a sequence of FVI iterates with constant regression error ε such that $\limsup_{k\to\infty} \|V^* - V_k\|_\infty = \varepsilon/(1-\gamma)$, showing the bound in Proposition 8.2 is tight.

Hint: Choose the regression error to always push V_k in the wrong direction by exactly ε.

Exercise 9.4 (LSPI with Exact Features). $(**)$
Suppose the feature class is rich enough that $Q^{\pi_k} \in \{Q_w : w \in \mathbb{R}^d\}$ for all policies π_k. Show that LSPI reduces to exact PI and converges to π^* in finitely many iterations.

Exercise 9.5 (Comparing API and FVI Error Amplification). $(***)$
Consider a family of MDPs parameterized by $\gamma \in [0.9, 0.999]$, each with the same per-iteration approximation error $\varepsilon = 0.01$.

(a) Plot the API bound $2\gamma\varepsilon/(1-\gamma)^2$ and the FVI bound $\varepsilon/(1-\gamma)$ as functions of γ.
(b) At what value of γ does the API bound first exceed 100? The FVI bound?
(c) Discuss the implications for choosing between API and FVI in long-horizon problems.

10 Summary

- **Approximate Policy Iteration** generalizes exact PI by allowing errors $\varepsilon_{\text{eval}}$ in evaluation and $\varepsilon_{\text{improve}}$ in improvement.
- The **error propagation bound** shows $\limsup_k \|V^* - V^{\pi_k}\|_\infty \leq \frac{2\gamma(\varepsilon_{\text{eval}}+\varepsilon_{\text{improve}})}{(1-\gamma)^2}$: errors are amplified by $1/(1-\gamma)^2$, making long-horizon problems fundamentally harder.
- Achieving ε-optimality requires per-iteration accuracy $O(\varepsilon(1-\gamma)^2)$, which becomes extremely stringent for γ close to 1.
- **LSTD** computes the TD fixed point w_{TD} directly by solving $\hat{A}w = \hat{b}$, avoiding the step-size tuning of semi-gradient TD at the cost of $O(d^2)$ per-step and $O(d^3)$ solve complexity.
- **LSPI** combines LSTDQ (LSTD for action values) with greedy improvement, forming a complete batch RL algorithm. It reuses a fixed data set across PI iterations, making it data-efficient but requiring adequate state–action coverage.
- **Fitted Value Iteration** approximates the Bellman optimality operator with supervised learning and achieves a better error amplification factor of $1/(1-\gamma)$, but requires computing the max over actions.

- All approximate methods solve the same fundamental tradeoff: richer function classes reduce approximation error $\varepsilon_{\text{eval}}$ but increase estimation error (variance) from finite samples.

11 Further Reading

- **Lesson 53** examines what happens when the evaluation step uses off-policy data with function approximation and bootstrapping: the *deadly triad* can cause LSTD-like methods to diverge. GTD2 resolves this with a true gradient objective.
- Bertsekas and Tsitsiklis [1996, Chapters 5–6] provide the most comprehensive treatment of approximate dynamic programming error bounds, including extensions to stochastic shortest path problems.
- Szepesvári [2010, Chapter 3] gives finite-sample bounds for LSTD and LSPI, showing that $O(d/(1-\gamma)^2)$ samples suffice for ε-accurate evaluation.
- For the connection to modern actor-critic methods, see the policy gradient lessons (Lesson 54 and Phase 04). Actor-critic algorithms are approximate PI where the "critic" performs approximate evaluation (often via semi-gradient TD) and the "actor" performs approximate improvement (via policy gradient).
- Bertsekas [2009] develops projected equation methods that unify LSTD, LSPE (Least-Squares Policy Evaluation), and related algorithms under a variational framework.

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